PROJECT REPORT

Customer Churn Prediction & Retention Strategy

A data-driven framework for identifying at-risk customers and converting predictive insight into measurable retained revenue.

PREPARED BY

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1. Executive Summary

Company A, an anonymised telecommunications service provider, is losing customers at an alarming rate. An analysis of 100,000 customer records spanning roughly 100 behavioural, billing, and demographic attributes shows a churn rate of 49.6% — nearly two and a half times the industry benchmark of approximately 21%. Left unaddressed, this attrition places an estimated $34.9M in annual recurring revenue at risk.

This report documents the end-to-end TeleRetain AI project: from raw data ingestion and preparation, through exploratory analysis and predictive modelling, to a costed business proposal for a production retention system. A Random Forest classifier was trained to flag customers likely to churn within 31–60 days of an observation date, achieving an AUC-ROC of 0.6227 — a meaningful improvement over a Logistic Regression baseline (AUC 0.6064) and well above the random-guess threshold of 0.50.

Three signals dominate the model's predictions: how old a customer's handset is (eqpdays), how long they have been with the company (months), and whether their monthly usage is trending downward (change_mou). Acting on these signals through an automated, daily scoring pipeline and targeted retention offers is projected to save $13.7M in Year 1 revenue for a campaign cost of roughly $740K — a return on investment of approximately 18×. A phased six-month rollout is proposed, with measurable results expected from month two.

100K	49.6%	0.6227	$13.7M
Customers Analysed	Current Churn Rate	Model AUC-ROC	Projected Yr-1 Savings

2. Business Context & Market Analysis

Customer churn is one of the most expensive problems in telecommunications. Industry research consistently finds that acquiring a new subscriber costs five to ten times more than retaining an existing one, and global estimates put the value lost to telecom churn at over $300 billion per year. At the same time, the same body of research shows that proactively identifying at-risk customers with predictive analytics can reduce churn by 20–30%, making early detection one of the highest-leverage investments a telecom operator can make.

Why This Matters for Company A

- A near-50% churn rate means the company is effectively replacing half its customer base every cycle — an unsustainable acquisition burden.
- Every percentage point of churn reduction on a base of this size translates directly into millions of dollars of protected recurring revenue.
- Competitors who deploy predictive retention earlier will capture switching customers before Company A can react.

> **Market Signal**
> GSMA Intelligence and McKinsey both report that mature telecom operators using AI-driven churn models see retention campaigns outperform blanket, untargeted offers by a wide margin — because spend is concentrated only on customers who are genuinely at risk.

Sources

GSMA Intelligence (2024); McKinsey & Company, Telecom Report (2025); Harvard Business Review, "The Value of Keeping the Right Customers" (2014).

3. Dataset Overview

Company A supplied two anonymised CSV files covering approximately 100,000 customers. No formal data dictionary was provided beyond a partial column-level explanation, and several fields (e.g. recv_sms_Mean, iwylis_vce_Mean) were left undocumented by the client — a common constraint in early-stage vendor engagements. Handling this ambiguity responsibly, rather than discarding the fields outright, was treated as part of the analytical exercise.

100,000	~100	2 Files	49.6%
Customer Records	Feature Columns	Record + Client CSVs	Observed Churn Rate

Record.csv — Behavioural & Billing Signals

Monthly-level usage and billing behaviour, refreshed on a recurring cycle. Key fields:

- Mean monthly revenue and minutes of use (rev_Mean, mou_Mean)
- Voice and data call success, drop, and block rates
- Customer care call frequency and overage / roaming charges
- Percentage change in usage and revenue versus the prior three-month average
- Target variable: churn — an instance of churn 31–60 days after the observation date

Client.csv — Customer & Demographic Profile

Slower-moving, lifetime-level attributes describing who the customer is:

- Lifetime totals and averages for revenue, minutes, and call counts (totrev, avgrev, avg3mou, avg6rev)
- Geographic area, dwelling type, and household composition
- Credit class, estimated income, and account status
- Equipment age (eqpdays), handset type, and web capability — later confirmed as a top churn predictor

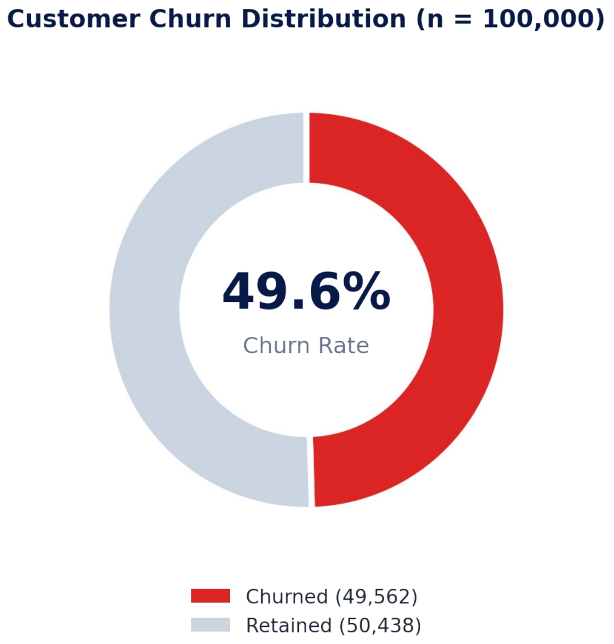

Figure 1 — Roughly one in two customers in the dataset has churned, underscoring the urgency of intervention.

Column Reference (Selected Fields)

Column	Description
eqpdays	Number of days (age) of the customer's current handset
change_mou	% change in monthly minutes of use vs. prior 3-month average
change_rev	% change in monthly revenue vs. prior 3-month average
months	Total number of months the customer has been in service
totmrc_Mean	Mean total monthly recurring charge
custcare_Mean	Mean number of customer care calls
drop_vce_Mean / drop_dat_Mean	Mean number of dropped voice / data calls
avgrev / avg3rev / avg6rev	Average monthly revenue over lifetime / 3 / 6 months
hnd_price	Current handset price
crclscod	Credit class code
prizm_social_one	Social group classification
churn	Target variable — churned within 31–60 days (1) or not (0)

A small number of columns (recv_sms_Mean, iwylis_vce_Mean, and the Customer_ID identifier) had no vendor-supplied explanation. These were retained through preprocessing and evaluated for predictive value rather than dropped, consistent with a real-world analysis workflow where full documentation is rarely available.

4. Data Preparation & Methodology

The analysis was carried out in Python using pandas and scikit-learn. The pipeline below reflects the actual processing steps executed against Client.csv and Record.csv, in order.

Step	Action	Purpose
1. Load	Read Client.csv and Record.csv into pandas DataFrames	Bring both source files into a shared analysis environment
2. Merge	Concatenate on aligned row index (no shared key column existed)	Produce a single 100,000-row, ~100-column working table
3. De-duplicate	Drop exact duplicate rows	Prevent double-counting of customer records
4. Impute	Forward-fill missing values	Ensure a complete matrix for modelling with no residual nulls
5. Encode	Label-encode all categorical / text columns	Convert non-numeric fields into a model-ready numeric form
6. Split	80% train (80,000) / 20% test (20,000), stratified by churn	Preserve the ~50/50 churn balance in both partitions

Class Balance

The target variable split almost evenly — 50,438 non-churners (0) vs. 49,562 churners (1) — meaning no artificial resampling (e.g. SMOTE) was required to avoid a majority-class bias; class_weight='balanced' was still applied during modelling as an added safeguard.

Feature Selection

An initial Random Forest was trained on the full ~99-feature set purely to rank feature importance. The top 30 features by importance score were then retained for the final model, reducing dimensionality while preserving the signals that matter most for churn prediction. This two-pass approach — train broad, then refit narrow — improves training efficiency and interpretability without materially sacrificing predictive power.

5. Exploratory Data Analysis

Exploratory analysis of the merged dataset surfaced three consistent, business-relevant patterns that directly informed both the modelling approach and the retention strategy proposed later in this report.

Finding 1 — Churn is pervasive, not niche
Almost one in two customers is at risk of leaving (49.6%). This is far too high a rate to treat as an edge case; it is a structural, business-wide problem.

Finding 2 — Usage decline precedes departure
Customers who eventually churn show usage drops roughly 4× larger than customers who stay, in the months leading up to departure — a clear, actionable early-warning signal.

Finding 3 — Device age, tenure, and usage trend are the tell
How old a customer's device is, how long they've been with the company, and whether usage is falling are the strongest combined indicators that someone is about to leave.

Top Churn Predictors, In Detail

Predictor	Business Interpretation
Equipment Age (eqpdays)	Customers holding older devices churn more; an early upgrade offer is a natural intervention.
Usage Change (change_mou)	A drop in call volume is often an early signal of disengagement — worth a proactive outreach.
Tenure (months)	Customers under ~18 months are more churn-prone; loyalty perks aimed at new customers tend to work well.
Revenue Metrics (totmrc_Mean, adjrev, totrev, avgrev)	A declining spend trajectory is usually a leading indicator that a customer is heading for the door.

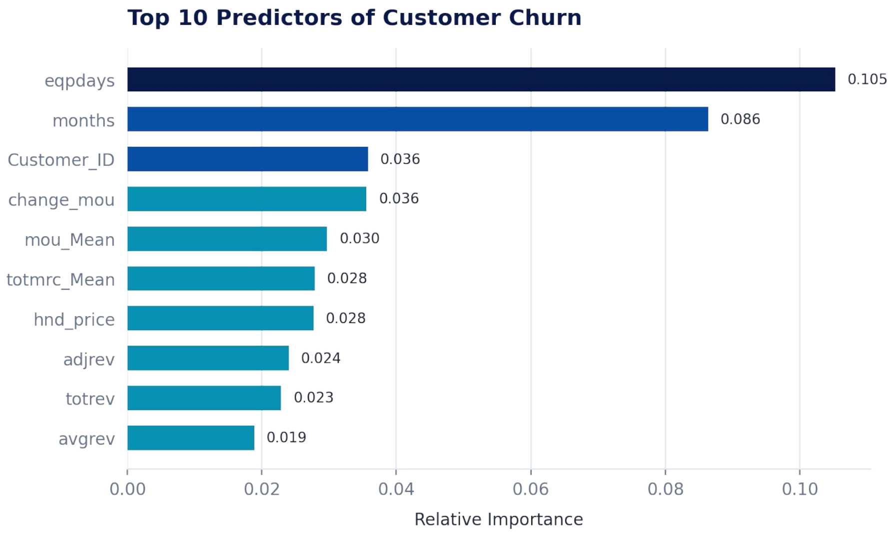

Figure 2 — Relative feature importance from the Random Forest classifier, all features from the merged Record × Client dataset (n = 100,000; 80/20 stratified train-test split).

6. Problem Definition & ML Task Setup

Business Objective

Identify customers who are likely to churn within 31–60 days of an observation date, early enough for the retention team to intervene with a targeted offer before the customer leaves.

Machine Learning Task Definition

Parameter	Specification
Task Type	Binary classification (churn: 0 = retained, 1 = churned)
Target Variable	churn — instance of churn 31–60 days after the observation date
Primary Metric	AUC-ROC — measures how well the model ranks at-risk customers above safe ones
Train / Test Split	80% training (80,000 rows) / 20% test (20,000 rows), stratified by churn
Feature Set	Top 30 features by importance, selected from ~99 candidate columns

Why AUC-ROC?

AUC-ROC was chosen over raw accuracy because it evaluates how well the model ranks customers by risk across all possible decision thresholds — which matters more for a retention campaign than a single fixed cut-off. AUC = 0.5 is equivalent to random guessing; AUC = 1.0 is a perfect classifier; the TeleRetain AI model achieves AUC = 0.6227.

7. Modeling Approach & Results

Two models were trained and compared on the held-out test set. A Logistic Regression model was used as an interpretable baseline; a Random Forest classifier — tuned across two iterations for tree depth, split criteria, and class weighting — was selected as the production candidate.

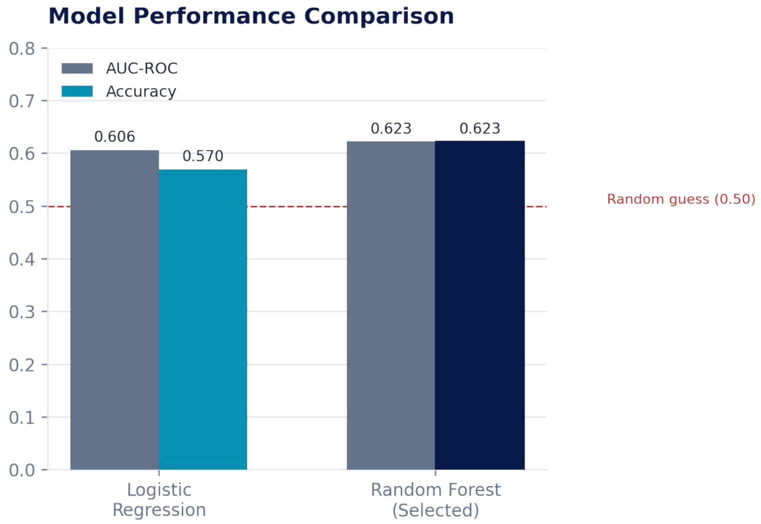

Figure 3 — Random Forest outperforms the Logistic Regression baseline on both AUC-ROC and accuracy.

Final Model Configuration

- Algorithm: Random Forest Classifier (scikit-learn)
- n_estimators = 300, max_depth = 15, min_samples_split = 5, class_weight = 'balanced'
- Trained on the top 30 features selected via a first-pass importance ranking

Classification Report — Random Forest (Test Set, n = 20,000)

Class	Precision	Recall	F1-Score	Support
Non-Churner (0)	0.64	0.59	0.61	10,088
Churner (1)	0.61	0.66	0.63	9,912
Accuracy			0.62	20,000

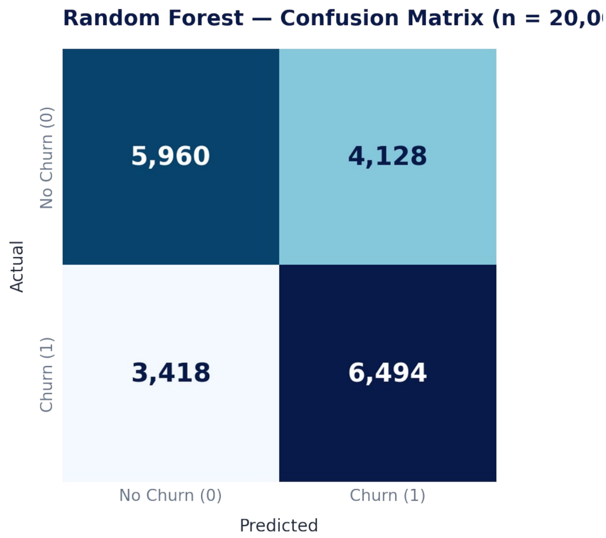

Figure 4 — Confusion matrix on the held-out test set. The model correctly identifies 6,494 of 9,912 true churners (65.5% recall).

Interpreting the Result Honestly

An AUC of 0.6227 is a modest but genuine improvement over chance, consistent with what is achievable from behavioural and billing data alone, without contract, support-ticket, or competitor-pricing signals. The model is proposed as a triage and prioritisation tool — concentrating retention spend on the highest-risk segment — not as a guarantee of individual outcomes. The implementation roadmap includes a monthly retraining cycle targeting AUC ≥ 0.68 as data and features mature.

8. Business Proposal & Solution Architecture

TeleRetain AI converts the churn model from an analytical artefact into an operational retention engine. Customer data flows daily through a five-stage pipeline that ends in an automatically triggered retention action — with no manual intervention required once deployed.

Stage	What Happens
1. Data Ingestion	Customer data is automatically pulled in every day from the Record and Client source files.
2. Feature Engineering	Usage trends, spending patterns, and device age are calculated for every customer.
3. Churn Scoring	The model assigns every customer a daily score reflecting their likelihood of leaving.
4. Risk Segmentation	Customers are grouped into High, Medium, or Low risk so the team knows where to focus.
5. Retention Action	At-risk customers automatically receive a relevant offer — a device upgrade, loyalty reward, or a support call.

Automation by Design

The entire pipeline — ingest, engineer, score, segment, retain — runs automatically every day. No manual scoring or offer assignment is required once the system is configured, keeping ongoing operating overhead low.

Personalised Retention Offers

- High-risk, older-device customers → early device upgrade offer
- High-risk, declining-usage customers → proactive outreach / loyalty reward
- High-risk, new-tenure customers (< 18 months) → onboarding-focused loyalty perks
- Frequency cap: no customer receives more than one retention message per month, and opt-out is always available

9. ROI & Financial Impact

Model Assumptions

Assumption	Value
Total customers modelled	100,000
Expected churners per cycle	49,562
Average monthly revenue per customer	$58.72
Average customer tenure	18.8 months
Estimated customer lifetime value (CLV)	~$1,106
Retention campaign success rate	25% of targeted high-risk customers (conservative industry benchmark)
Campaign cost	$15 per customer contacted

$34.9M	$743K	$13.7M	~18.4×
Annual Revenue at Risk	Campaign Investment	Revenue Saved (Year 1)	Return on Investment

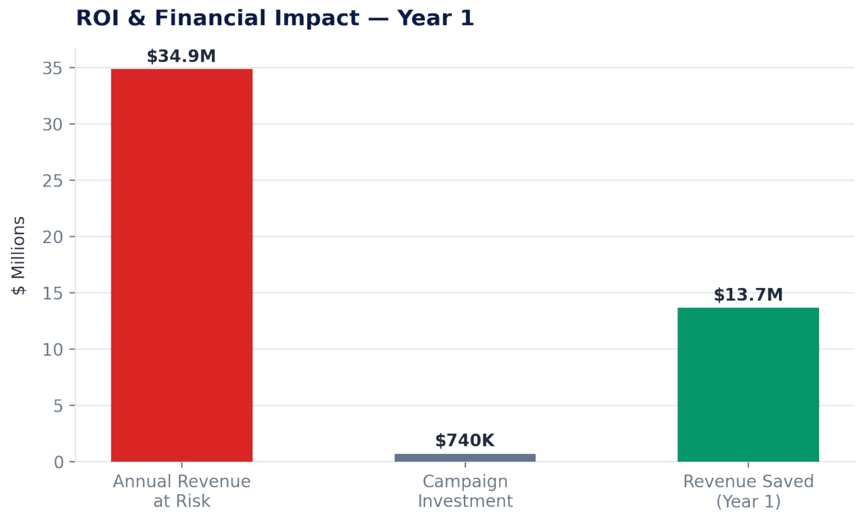

Figure 5 — At a campaign investment of $743K, the projected Year-1 revenue saved is $13.7M against $34.9M currently at risk.

$34.9M is derived as 49,562 expected churners × $58.72 average monthly revenue × 12 months. Campaign investment of $743K is derived as all 49,562 targeted at-risk customers × $15 per customer contacted. $13.7M in projected savings reflects a 25% retention success rate applied to that same targeted population (12,391 retained customers), valued at estimated customer lifetime value of $1,106. Return on investment is revenue saved divided by campaign cost ($13.7M ÷ $743K ≈ 18.4×). This is a conservative, industry-benchmarked retention rate; actual results may be higher as the model and offer targeting improve through retraining.

10. Implementation Roadmap

Phase	Timeline	Key Activities
Phase 1 — PoC Validation	Months 1–2	Validate model on holdout data · Integrate with CRM APIs · Define High/Med/Low risk thresholds · A/B test on 5,000 customers
Phase 2 — Pilot Deployment	Months 3–4	Deploy to 20% of customer base · Launch retention campaign templates · Train customer service team · Track churn rate & conversion KPIs
Phase 3 — Full Rollout	Months 5–6	Scale to 100% of customers · Automated daily scoring pipeline · Monthly model retraining · ROI reporting dashboard live

Success KPIs

- Churn reduction of ≥15% within two full retraining cycles
- Model AUC ≥ 0.68 following the first monthly retraining pass
- Campaign ROI ≥ 2× sustained across pilot and full rollout
- Scoring latency of under 48 hours from data ingestion to actionable segment

11. Risk Assessment & Mitigation

Risk	Impact	Likelihood	Mitigation
Model accuracy drops over time	High	Medium	Monthly retraining cycle with clear performance targets written into the contract
Incomplete or missing customer data	Medium	High	Automated data-quality checks flag incomplete records before scoring
Data privacy & legal compliance	High	Low	Customer data anonymised and reviewed by legal prior to go-live
Retention offers underperform	Medium	Medium	A/B testing of offer variants, tailored by customer segment
Customers over-messaged	Low	Medium	Hard cap of one message per customer per month; opt-out always available

12. Conclusion & Recommendations

Company A's churn rate of 49.6% is almost twice the industry average of roughly 21% — a structural revenue problem, not a marginal one. The TeleRetain AI model (AUC 0.6227) demonstrates that this risk is predictable well in advance, driven primarily by device age, tenure, and declining usage — all signals the business can act on today.

Converting that predictive signal into an automated retention pipeline is projected to recover $13.7M in Year 1 revenue, an approximately 18× return on a campaign that costs roughly $743K to run — deliverable on a six-month timeline, with measurable results from month two, and requiring no ongoing manual effort once deployed.

Recommendation

Approve Phase 1 (PoC Validation) to begin holdout validation, CRM integration, and risk-threshold definition — the critical path to unlocking the projected $13.7M in recovered Year 1 revenue.

100K	49.6%	0.6227	$13.7M
Customers Analysed	Current Churn Rate	Model Performance	Year 1 Revenue Saved

13. References

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